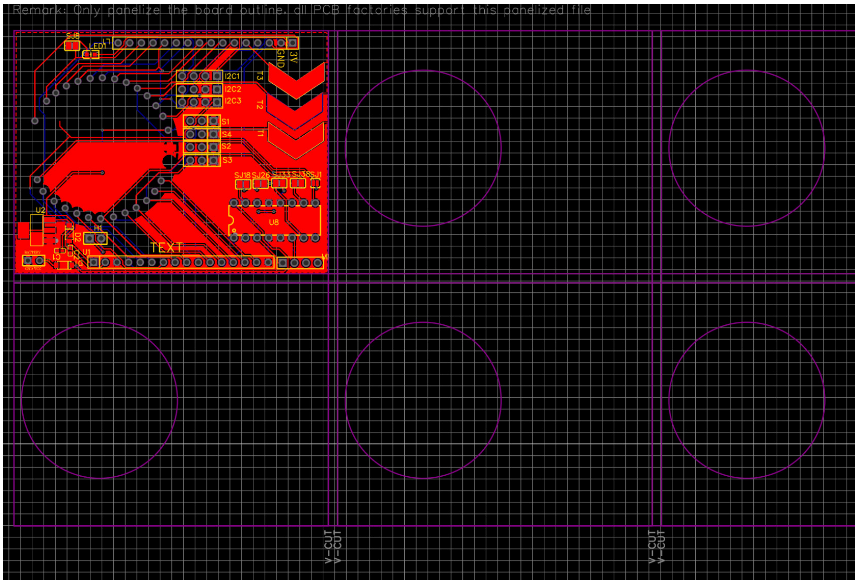

Fig. 6: V cut for rectangular PCBs

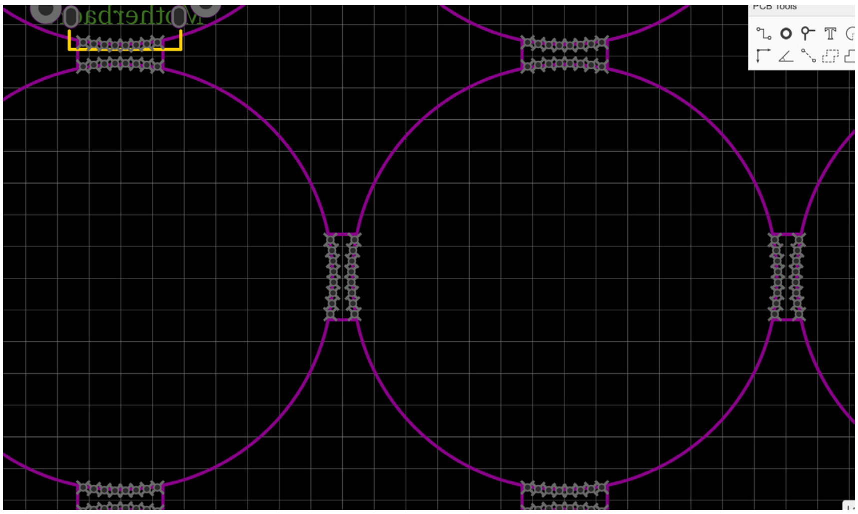

Fig. 7: Stamp hole based separation for irregular shaped PCBs

elisation. PCB manufacturers allow panels up to a specific size. Slight increase beyond this limit can incur the cost of an additional panel. For example, a panel with side rails that are each 10mm wide can be modified by removing the rails, freeing up 20mm of horizontal space. This space can then be used to add another column of PCBs, significantly increasing the number of units per panel. Here's an example.

1. Current layout:
   - Panel with side rails: 6x3 PCBs
   - Total PCBs: 18
2. Optimised layout:
   - Panel without side rails: 7x3 PCBs (or potentially 6x4 if vertical space allows)
   - Total PCBs: 21 (or 24)

Other methods and consideration to save cost

In addition to panelisation techniques, several other strategies can help cut PCB manufacturing costs. By optimising the depanelisation process and maximising panel efficiency, you can significantly reduce expenses without compromising quality. Consider following methods:

Depanelisation optimisation. V-scoring, which involves shallow cuts on both sides of the PCB, allows the boards to be snapped apart easily. This method is generally cheaper than tab routing as it requires no space for tabs and permits higher panel utilisation. However, V-scoring is only suitable for straight-line cuts and simpler shapes. Tab routing, on the other hand, is ideal for complex shapes and provides better handling during assembly, although it slightly reduces panel efficiency due to the space required for tabs.

Minimising wasted space. An optimised board layout arranges PCBs to minimise unused space. Techniques like rotating and strategically placing boards can maximise the number of boards per panel, reducing material waste and overall cost. Tools such as KiCad, Eagle, or Altium Designer can automatically optimise these layouts. Placing similar or compatible designs edge-to-edge also reduces gaps between boards, lowering the amount of cutting needed, saving both machine time and material.

Material and process choices. Choosing cost-effective materials, such as standard FR-4, and selecting the appropriate thickness to balance performance and cost, can result in substantial savings. For surface finishes, HASL (hot air solder levelling) is a common, cost-effective option, while ENIG (electroless nickel immersion gold) is more expensive but necessary for applications requiring better surface quality and durability.

This streamlined approach to panelisation and design not only reduces production costs but also optimises manufacturing efficiency. EFY

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